SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 **Total Marks:20**

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

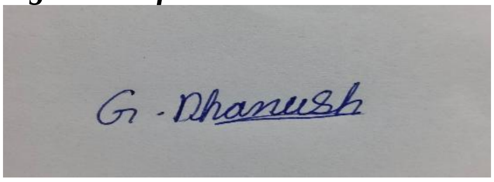
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I _Dhanush.G___(192512132)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



<u>RUBRICS FOR CALCULATION:</u>

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

1.

ANSWER:

Smart Home Automation System Using Intelligent Agents

Introduction

A smart home automation system controls lighting, temperature, and security by using information from sensors such as motion detectors, temperature sensors, light sensors, and security cameras. The system should respond to real-time events while learning the habits and preferences of the users. Artificial Intelligence (AI) agents help make intelligent decisions in such environments.

Types of Intelligent Agents

1. Simple Reflex Agent

A Simple Reflex Agent acts only on the current input using predefined condition-action rules. It does not remember previous events.

Working in Smart Home:

- If motion is detected → Turn ON the lights.
- If no motion for 10 minutes → Turn OFF the lights.
- If temperature > 30°C → Switch ON the fan or AC.
- If smoke is detected → Trigger the fire alarm.

Advantages

- Fast response.
- Easy to implement.
- Low computational cost.

Disadvantages

- Cannot learn user habits.
- Cannot handle complex situations.
- Makes decisions only from current sensor data.

2. Model-Based Reflex Agent

A Model-Based Agent keeps an internal model (memory) of the environment. It considers both the current sensor input and previous states.

Working in Smart Home

- Remembers whether lights are already ON.
- Detects whether someone is inside or outside the house.
- Uses weather information along with indoor temperature.
- Avoids unnecessary switching of appliances.

Advantages

- Better decision-making.
- Handles partially observable environments.
- More reliable than simple reflex agents.

Disadvantages

- Requires memory.
- Slightly more complex.

3. Goal-Based Agent

A Goal-Based Agent selects actions that help achieve specific goals.

Example Goals

- Maintain room temperature at 24°C.
- Keep the home secure.
- Minimize electricity consumption.

Working

If the room temperature rises above the desired value:

1. Check windows.
2. Close windows if open.
3. Turn ON the AC.
4. Monitor until the desired temperature is reached.

Advantages

- Plans future actions.
- Can compare multiple possible solutions.
- Flexible.

Disadvantages

- Requires search algorithms.
- Higher computational cost.

4. Utility-Based Agent

A Utility-Based Agent selects the action that provides the maximum overall benefit (utility). The utility function considers:

- Comfort
- Energy consumption
- Security
- Cost

Working

Suppose:

- User prefers 24°C.
- Electricity price is high.
- Outdoor weather is cool.

Instead of turning ON the AC, the agent opens the windows because it gives better comfort with lower electricity cost.

Advantages

- Produces optimal decisions.
- Balances multiple objectives.
- Handles conflicting goals.

Disadvantages

- Utility function design is complex.
- Requires more computation.

Comparison of Intelligent Agents

Agent Type	Memory	Learning	Planning	Best Use
Simple Reflex	No	No	No	Immediate responses
Model-Based	Yes	Limited	Limited	Tracking environment
Goal-Based	Yes	Possible	Yes	Goal achievement
Utility-Based	Yes	Yes	Yes	Best overall decision

Hybrid Approach

A hybrid system combines the strengths of multiple agents.

Reflex Agent

- Immediate emergency response
- Fire alarm
- Intruder detection

Model-Based Agent

- Tracks occupancy
- Maintains home status

Goal-Based Agent

- Plans heating/cooling schedules
- Optimizes lighting

Utility-Based Agent

- Minimizes electricity bill
- Maximizes user comfort
- Improves security

Justification

A **Hybrid Agent** is the most effective because:

- Responds immediately to emergencies.
- Learns user preferences over time.
- Plans future actions.
- Optimizes comfort, security, and energy efficiency simultaneously.

2.

ANSWER:

Robot Navigation in an Unknown Maze

Introduction

A robot is placed inside an unknown maze and must find the exit without knowing the map beforehand. The robot explores the maze using search algorithms and intelligent agent techniques.

Uninformed Search Strategies

Uninformed search algorithms explore the search space without heuristic knowledge.

1. Uniform Cost Search (UCS)

Principle

Uniform Cost Search expands the node with the lowest cumulative path cost.

Steps

1. Start from the initial position.
2. Store nodes in a priority queue.
3. Expand the node with minimum path cost.
4. Continue until the exit is found.

Example

```
Start

 /   \
A(2) B(4)

A
|
C(3)

Goal
```

Expansion order:

Start

↓

A (Cost=2)

↓

C (Cost=5)

↓

Goal

Advantages

- Finds the least-cost path.
- Complete.
- Optimal.

Disadvantages

- High memory usage.
- Slow for large mazes.

Complexity

Time Complexity:
$O(b^{(1+C^*/\varepsilon)})$
Space Complexity:
$O(b^{(1+C^*/\varepsilon)})$
where:

- b = branching factor
- C* = optimal path cost
- ε = minimum step cost

Advantages

- Very low memory requirement.
- Complete.
- Suitable for unknown depth.

Disadvantages

- Repeats node expansion.
- Slightly slower than BFS.

Complexity

Time Complexity
$O(b^d)$
Space Complexity
$O(bd)$
where:

- b = branching factor
- d = solution depth

Comparison of UCS and IDS

Feature	**UCS**	**IDS**
Complete	Yes	Yes
Optimal	Yes	Only with equal step cost
Time Complexity	$O(b^{(1+C^*/\varepsilon)})$	$O(b^d)$
Space Complexity	$O(b^{(1+C^*/\varepsilon)})$	$O(bd)$
Memory Usage	High	Low
Best Use	Different path costs	Unknown search depth

Relation to Intelligent Agents

Simple Reflex Agent

- Moves based only on immediate surroundings.
- Cannot remember visited locations.
- Often gets trapped in loops.

Model-Based Agent

- Maintains an internal map.
- Remembers explored paths.
- Avoids revisiting the same locations.

Goal-Based Agent

- Goal: Reach the exit.
- Plans exploration using search algorithms.
- Chooses the best available path.

Utility-Based Agent

Evaluates:

- Distance
- Energy consumption
- Safety
- Time

Chooses the action with the highest utility.

Effect of Agent Design on Decision Making

Agent	**Decision Style**
Simple Reflex	Immediate reaction
Model-Based	Uses memory
Goal-Based	Searches toward goal
Utility-Based	Optimizes multiple factors

Most Effective Combination

Recommended Agent

Model-Based Goal-Based Agent

Recommended Search Strategy

- **Uniform Cost Search (UCS)** if movement costs differ (e.g., rough terrain, slippery paths).
- **Iterative Deepening Search (IDS)** if all moves have equal cost and memory is limited.

Justification

A **Model-Based Goal-Based Agent** remembers explored areas, builds knowledge of the maze, and plans routes to the exit. Pairing it with **UCS** guarantees the least-cost path when costs vary, while **IDS** is more memory-efficient when all moves cost the same. This combination improves navigation efficiency, avoids repeated exploration, and ensures reliable decision-making.